

Low Voltage Variable Speed Drive

1 General

- Furnish complete variable frequency drives as specified herein for the pumps designated to be variable speed. All standards and optional features shall be included within the VFD enclosure. The drive panel shall be free-stand with IP 54.
- VFD is to be provided for motor starting and speed control where shown on electrical or mechanical drawings. Necessary interfaces and accessories to be provided for communication with the SCADA system.
- VFD shall convert three-phase, 50 hertz utility power to adjustable voltage and frequency, three phase power for stepless motor speed control from 10% to 100% of the motor's 50 hertz speed. Input voltage shall be as specified on the drawing schedules.
- The VFD shall be with built-in active harmonics filter at the rectifier & inverter units (**Low Harmonic Type**). The harmonic filter configuration shall result in a current waveform that approximates near sinusoidal input current waveform and compliant with requirements of IEEE 519. The power section shall be insensitive to phase rotation of the AC line. Total harmonic voltage distortion less than 3%.
- VFD shall be type tested, certification should be submitted.
- Enclosure should be fully closed metal casing shall be made of four pre-lacquered, hot-galvanized steel plate sections 2 mm thick to optimize heat exchange.
- The output power section shall change fixed DC voltage to adjustable frequency AC voltage. This section shall use insulated gate bipolar transistors (IGBT) or intelligent power modules (IPM) as required by the current rating of the motor.
- VFD shall include a converter and an inverter section. The converter section shall convert fixed frequency and voltage AC utility power to DC voltage. All VFDs shall include input line reactors or an isolation transformer as required. The inverter section of the VFD shall invert the DC voltage into a quality output waveform, with adjustable voltage and frequency for stepless motor speed control. The VFD shall maintain a constant V/Hz ratio. The VFD and options shall be tested to ANSI/UL.508. The complete drive, including all specified options, shall be listed by a nationally recognized testing agency such as UL or ETL.
- Power line noise shall be limited to a voltage distortion factor and line notch as defined in IEEE 519, "Recommended Practices and requirements for harmonic control in electrical power

systems". The total voltage distortion shall not exceed 3 percent. The total feedback harmonics not to exceed 10%. 

- The VFD shall not emit radiated RFI in excess of the limitations set forth in the FCC Rules and Regulations, FCC Part 15 for Class A computing devices. The VFD shall carry a FCC compliance label. PWM type drives shall include RFI filters. 
- Motor noise as a result of the VFD shall be limited to 3 dB over across the line operation measured at 1 meter from the motor's centerline. 
- VFD's full load amp rating shall meet or exceed NFPA 70, NEC Table 430-150. 
- Protective Features shall include:
 - a) Individual motor overload protection for each motor controlled.
 - b) Protection against input power under-voltage, over-voltage, and phase loss.
 - c) Protection against output current overload and instantaneous over-current.
 - d) Protection against over-temperature within the VFD enclosure.
 - e) Protection against over-voltage on the DC bus.
 - f) Thermistor motor protection relay interface. 
 - g) Protect VFD from sustained power of phase loss. Under-voltage trip activates automatically when line voltage drops more than 10 percent below rated input voltage.
 - h) Automatically reset faults due to under-voltage, over-voltage, phase loss, or over temperature.
 - i) 9. Protection against output short circuit and motor winding shorting to case faults, as defined by UL 508.
 - j) Status lights or digital display for indication of individual fault conditions.
 - k) Controller capable of operating without a motor or any other equipment connected to the drive output to facilitate start-up and troubleshooting.
 - l) Input line reactors shall be provided to minimize harmonics introduced to the AC line and to provide additional protection to AC line transients.
- Interface Features for the VFD will include:
 - a) Door mounted Hand/Off/Auto selector switch to start and stop the VFD. In the Auto position, the VFD will start/stop from the controller. In the Hand position, the VFD will run regardless of the remote contact position.
 - b) Manual speed control capability 
 - c) Local/Remote selector switch. In the Remote position, motor speed is determined by the follower signal. In the local position, motor speed is determined by the manual speed control. 

- d) Power/On light to indicate that the VFD has tripped on a fault condition.
- e) Digital meter with selector switch to indicate percent speed and percent load.
- f) A set of form-C dry contacts, to indicate when the VFD is in the Run mode.
- g) A set of form-C dry contacts, to indicate when the VFD is in the Fault mode.
- h) A 0 to 10 V output signal to vary in direct proportion to the controller's speed.
- i) VFD to have terminal strip and N.C. safety contacts such as freezstats, smoke alarms, etc.
VFD to safety shut down in drive or bypass mode when contacts open.
- Adjustments will include the following:
 - a) Maximum speed, adjustable 15 to 100 percent base speed.
 - b) Acceleration time, adjustable 3 to 60 seconds.
 - c) Deceleration time, adjustable 3 to 60 seconds with override circuit to prevent nuisance trips if decal time is set too short.
 - d) Current limit, adjustable 0 to 105 percent.
 - e) Overload trip set point.
 - f) Offset and gain to condition the input speed signal.
- Service conditions will cover:
 - g) Ambient temperature, 0 to 45 deg C.
 - h) 0 to 85 percent relative humidity, non-condensing.
 - i) Elevation to 1,000 meters without derating.
 - j) AC line voltage variation, -10 to +10 percent of nominal
- Special features will include the following features in the VFD enclosure while maintaining its UL to ETL listing:
 - k) Manual by-pass shall provide all the circuitry necessary to transfer the motor from the VFD to the power line or from the line to the controller. The by-pass circuitry shall be mounted in a separate section of the VFD enclosure. Motor overload protection shall be provided in both drive and by-pass modes.
 - l) A door interlocked, pad lockable drive disconnect switch shall be provided to disconnect power from the VFD only (The disconnect shall be clearly marked as such).
 - m) A motor circuit protector, MCP, shall be provided as a means of disconnecting all power to both the VFD and by-pass circuits as well as providing short circuit and locked rotor protection to the motor while in the by-pass mode.

- n) The disconnect and by-pass functions may be accomplished via disconnects, contractors and overloads, or with a four position Drive/Off/Line/Test switch with motor starter and by-pass fuses. 

2 Required signals for control system

- Inputs and outputs

At least, the following standard Inputs and Outputs shall be provided, to be used in interface with the control system:

Analogue Inputs: 1 x Programmable current input 0(4) - 20mA

Analogue Output: 1 x Programmable analogue outputs 0(4) - 20mA

Logic inputs: 6 x Programmable logic Inputs isolated from the mains 

(One of these inputs could be used for PTC probe)

Specific Fault, local, off, remote

Safety input one input dedicated to the Power removal safety function

- Communications

The AC drive shall have the capability for internal mounted communication card. The Modbusplus, Profibus DP, DeviceNet protocols shall be the minimum available for combatable with the control system. 

Operating conditions:

- Operating ambient Temperature: up to 50 °C 
- Max. Relative Humidity (IEC 60068-2-3): 5...95 %, without condensation 
- Max. Corrosion Level of the Cooling Air: IEC 721-3-3, class 3C1.

3 Submittals

The contractor shall submit the flowing:

- Technical Data: Include dimensions, ratings and data on features and components.
- For Variable Frequency Drives: Submit manufacturer's performance data including dimensional drawings, customer connection drawings, power circuit diagrams, installation and maintenance manuals, warranty description, VFDs FLA rating, certification agency file numbers and catalogue information.
- Point-wise compliance statement to the specifications, duly signed by the manufacturer / manufacturer's authorized representative and the contractor.
- Shop and Installation Drawings: Submit for each motor control center, and prior to ordering materials and equipment, specified in this Section. Include dimensioned plans, elevations,

and component lists. Show ratings, including short time and short circuit ratings, and horizontal and vertical bus ampacities.

- Complete technical schedule of features, characteristics, ratings, and factory settings on all motor starters, motor protection relays, sensing units, control accessories, etc.
- Instructional details of equipment, particularly motor control centers and panels.
- Installation details of motor control centers and panels and of control and sensing accessories.
- Exact routing of power and control cables, wiring and conduits.
- Feeder termination details at motor control centers, starters, motors, isolating switches, control and sensing accessories etc.
- Wiring Diagrams: Interconnecting wiring diagrams (of all power and control circuits) pertinent to class and type specified for motor control center. Schematic diagram of each type of controller unit indicated.
- Wiring Diagrams: Submit for approval detailed control wiring diagrams and a list of control equipment with descriptive literature.
- Free hand field wiring diagrams or sketches will not be accepted.
- Maintenance Data: For products to include in the maintenance manuals as specified
- Submit certified protection relay coordination settings for review, along with certified "Asbuilt" connection drawings, before energisation of equipment.
- As-built (Record) Drawings: At project close-out, submit certified record drawings of installed electrical motor control centers, in accordance with requirements of the specifications, Division 1.

4 Coordination

- Coordinate features of controllers and accessory devices with pilot devices and control circuits to which they connect.
- Coordinate features, accessories, and functions of each motor controller with the ratings and characteristics of the supply circuit, the motor, the required control sequence, and the duty cycle of the motor and load.

POWER TRANSFORMER

1. General

Transformers shall be of the oil immersed, self cooled, double wound, core type suitable for indoor floor mounting. They shall be designed to withstand tropical conditions and shall give efficient service with minimum maintenance. Transformers shall be constructed in accordance with the specifications and regulations of the country where manufactured except for temperature rise which shall not exceed 60° C for windings measured by resistance and 55° C for the top of oil measured by thermometer and must not reach a value that will injure the core itself or adjacent material when the transformer carries its rated output continuously at 0.8 power factor and ambient temperature of 45° C. Two English copies of transformer specifications should be submitted with tenders.

The neutral point of the L.T. side of the transformer shall be brought out to a separate terminal which shall be earthed. The transformer shall be low losses type, F/B insulation class and temp rise.

2. Primary and Secondary Voltage Levels

The primary voltage shall be 22KV (+5%, - 10%) and secondary 400-230V, and as indicated in the single line diagram.

3. Impedance Voltage

The short circuit impedance of the transformer shall be 4 - 7 %.

4. Tapping and Tap Changing

The high voltage coils shall be provided with tappings to allow variation in the supply voltage of +2*2.5%, -4*2.5% of nominal voltage. The tapping change device shall be of the "No Load" type and outside the tank mounted.